Global Facilities - Design & Construction Standard - Mechanical - Probe Cleanroom HVAC System

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Probe Cleanroom HVAC Systems for Front End Sites as part of new Greenfield FAB Construction projects.

1.2 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 The document covers Probe Cleanroom HVAC Systems for Front End Sites including airflow supply, conditioning, and recirculation for control of Cleanroom particles, temperature, humidity, and pressurization. It does not cover other Cleanroom Systems such as Civil, Architectural, Structural (CSA), Fire Suppression, Electrical, Controls, or other Cleanroom Utilities.

2.2 Site(s) impacted

2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
AMC	Airborne Molecular Contaminants
CHW	Chilled Water
CD	Condensate Drain
DB	Dry Bulb
DCC	Dry Cooling Coil
DP	Dewpoint
FAT	Factory Acceptance Test
FFU	Fan Filter Unit
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
HEPA	High Efficiency Particulate Arrest (Filter)
ICA	Instrument Control Air
IST	Integrated System Test
ITP	Inspection and Test Plan
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
MAU	Makeup Air Unit
N+1+F	System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +1 or more future units
OAT	Operational Acceptance Test
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
RA	Return Air
RAC	Return Air Chase
RH	Relative Humidity
SA	Supply Air
SAT	Site Acceptance Test
TUM	Tool Utility Matrix
ULPA	Ultra Low Penetration Air (Filter)
UPS	Uninterruptible Power Supply
VSD/VFD	Variable Speed Drive / Variable Frequency Drive
WB	Wet Bulb
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities

Term/Acronym	Definition / Description
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

5.1 The following systems are used to support the Cleanroom HVAC:

5.1.1 **Cleanroom Recirculation Air** is the main Cleanroom air that is recirculated to provide control of Cleanroom dry bulb temperature, particle cleanliness levels, and AMC levels. This air is conditioned via the Dry Cooling Coils, filtered via the FFU ULPA filters, and supplied via the FFU fans to the Cleanroom. Chemical Filters may be used to filter and remove AMCs.

The recirculation airflow in Cleanrooms is used to meet the cooling loads and also to meet the dilution requirement to reduce room particle concentration.

5.1.2 **Cleanroom Makeup Air** is 100% outside air that is supplied to provide control of Cleanroom humidity and pressurization levels. The air is conditioned, filtered, and supplied to the Cleanroom via the Makeup Air Units.

5.1.3 **Cleanroom Exhaust** is air that is removed from the Cleanroom for process/tool exhaust or general space ventilation.

5.2 The main Cleanroom HVAC equipment and components shall be as follows:

5.2.1 Fan Filter Units (FFU)
5.2.2 Dry Cooling Coils (DCC)
5.2.3 Return Air Chase
5.2.4 Return Air Plenum
5.2.5 Raised Floor System

5.3 Other Cleanroom Systems associated with Cleanroom HVAC include:

5.3.1 Makeup Air (MAU)
5.3.2 Exhaust

6.0 Cleanroom Configuration:

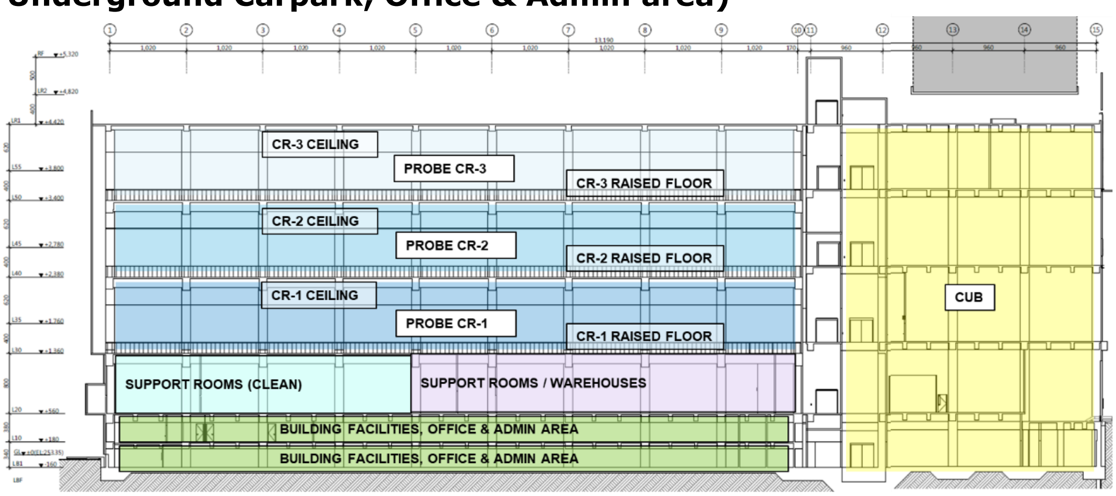
The following shall be the default Cleanroom Recirculation Air design concept for a typical Single Level Probe Cleanroom with 2 levels, consisting of 1 level of Probe Cleanroom and 1 level of Interstitial Plenum. The same concept can be applied to the building with multiple levels of Probe Cleanroom configuration such as a Double- / Triple- Level Probe Cleanroom.

An example of typical Triple-Level Probe Cleanroom and each level is shown in **Table 1 and Figure 1**. The number of levels of a Probe Cleanroom building will be project specific, but the Cleanroom Recirculation Air design concept shall be the same as outlined in this Standard.

Table 1: Typical Levels for Single and Multiple Level Probe Cleanroom

Typical Single Level Probe Cleanroom	Typical Triple Level Probe Cleanroom
Level 2 – Probe Cleanroom with Interstitial Plenum (Clean), & Raised Floor (Clean) Level 1 – Support Rooms / Warehouse (Clean) - (This Level is optional and project specific.)	Level 5 – Probe Cleanroom with Interstitial Plenum (Clean), & Raised Floor (Clean) Level 4 – Probe Cleanroom with Interstitial Plenum (Clean), & Raised Floor (Clean) Level 3 – Probe Cleanroom with Interstitial Plenum (Clean), & Raised Floor (Clean) Level 2 – Support Rooms / Warehouse (Clean) (This Level is optional and project specific.) Level 1 – Building Facilities, Office & Admin Area - (This Level is optional and project specific. It can be included or excluded)

Figure 1 – An example of Triple-Level Probe Cleanroom (3-Levels of Probe Cleanroom, 1-Level of Support Rooms, 2-Levels of Underground Carpark, Office & Admin area)



6.1 Cleanroom Configuration

The Cleanroom shall be designed as unidirectional airflow, with vertical downwards airflow pattern through the Probe Cleanroom.

Cleanroom Recirculation Air shall be drawn from the above-ceiling Interstitial Plenum by the Fan Filter Units, through the ceiling mounted FFU ULPA Filters, and supplied down through the Probe Cleanroom.

The Cleanroom air shall then pass through the Probe Cleanroom Raised Floor perforated or grated floor tile and then be routed into the Vertical Return Air Chase. The air will return to the above-ceiling interstitial plenum to complete the recirculation cycle.

At the Interstitial Plenum level, the Cleanroom Recirculation Air shall pass through the Dry Cooling Coils, where sensible heat is removed and the Cleanroom air's Dry Bulb Temperature is lowered to provide Cleanroom Temperature control.

Makeup air is supplied from the Makeup Air Units and mixed with the Cleanroom Recirculation Air, to provide the Cleanroom's Relative Humidity (and Dew Point Temperature) control. The Makeup air is also used to maintain the Cleanroom at positive pressure and compensate for the Exhaust being removed from the Cleanroom.

7.0 Cleanroom Specifications:

7.1 Cleanroom Classification and Particles

The Cleanroom HVAC System shall be designed and operated to meet the Cleanroom Classification Specifications listed in **Table 2**. This Classification based on particle size measured at 0.3μm. The Cleanroom Classification shall be initially certified at As-Built condition, but the Classification shall also apply through the entire life of the Cleanroom during Operational conditions.

Table 2: Probe Cleanroom and Auxiliary Cleanroom Space Specification - Particles

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	ISO 14644-1 Class 6	Probe Area (or Probe tool mini-environment)
Cleanroom Particles	ISO 14644-1 Class 6	AMHS Clean Link Bridge
Cleanroom Particles	ISO 14644-1 Class 7	Probe Gowning Room
Cleanroom Particles	ISO 14644-1 Class 7	Probe Decon Area and Equipment Move-in prior to Probe Cleanroom

7.2 Probe Cleanroom Temperature and Humidity

The Probe Cleanroom HVAC System shall be designed and operated to meet the Temperature and Humidity Specification listed in **Table 3**.

Table 3: Probe Cleanroom Specification – Temperature & Humidity

Parameter	Specification Requirement	Applicable Spaces
Cleanroom Temperature	22.5±2 °C [72.5±3.6 °F]	Probe Cleanroom
Cleanroom Humidity	Design Capability: 43±3 %RH Probe Spec: 45±5% RH	Probe Cleanroom

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Note: Although the design capability to achieve Cleanroom Humidity of 42% RH Setpoint is specified, the current Micron Global Spec Alignment recommendation for Cleanroom Humidity is 45% RH for Probe sites. Therefore, the Cleanroom HVAC system shall be designed with capability to meet 42% RH, but the sites can be operated at different setpoints to align with the Micron Global Spec Alignment.

Table 4 is a reference table of equivalent Dewpoint Temperatures, based on the Dry Bulb Temperature and Relative Humidity:

- Based on the Design Cleanroom Conditions of 22.5°C DB and 43% RH, the equivalent Dewpoint Temperature is 9.0°C DP; this applies to current design capacity requirement.

- Based on the Design Cleanroom Conditions of 22.5°C DB and 45% RH, the equivalent Dewpoint Temperature is 10.0°C DP; this applies to current Probe specification requirement.

Table 4: Cleanroom Environment - Equivalent Dewpoint Temperature (for Reference)

Dry Bulb Temp	Relative Humidity (% RH)							
	39%	**40%**	**42%**	**43%**	**45%**	**46%**	**47%**	**50%**
21.0°C [69.8°F]	6.6	6.9	7.6	8.0	8.7	9.0	9.3	10.2
21.5°C [70.7°F]	7.0	7.4	8.1	8.4	9.1	9.4	9.8	10.7
22.0°C [71.6°F]	7.5	7.8	8.5	8.9	9.6	9.9	10.2	11.1
22.5°C [72.5°F]	7.9	8.3	9.0	9.3	10.0	10.3	10.7	11.6
23.0°C [73.4°F]	8.3	8.7	9.4	9.8	10.5	10.8	11.1	12.1
23.5°C [74.3°F]	8.8	9.2	9.9	10.2	10.9	11.3	11.6	12.5
24.0°C [75.2°F]	9.2	9.6	10.3	10.7	11.4	11.7	12.0	13.0

7.3 Probe Cleanroom Pressurization

The Probe Cleanroom HVAC System shall be designed and operated to maintain the Pressurization Specification listed in **Table 5**.

Table 5: Probe Cleanroom Specification - Pressurization

Parameter	Design Requirement	Applicable Sites / Remarks
Cleanroom Pressurization	Minimum: 7-8 Pa Recommended: 10-12 Pa	Probe Cleanroom (Ballroom) **Note:** Pressurization is with reference to external (outdoor) environment

7.4 Cleanroom Airborne Molecular Contamination (AMC) (for Reference)

AMCs can cause product impact and yield loss due to corrosion, hazing, delaminating of films or plating, unwanted doping, etc. AMCs are typically classified as Acids, Bases, Condensable, and Dopants (ABCD). Examples of each type is listed in **Table 6**.

Table 6: Examples of Airborne Molecular Contamination Types

Acid	Base	Condensable /Organics	Dopants	No Classification
HF HCL NOx SOx HBr H2SO4	NH3 TMAH Amines	Siloxanes TMA DOP IPA Refractories TMS	B2H6 BF3	O3 H2O2 H2S

The Cleanroom HVAC System shall be designed and operated to meet the AMC Specifications listed in **Table 7**.

Table 7: Cleanroom Specification – AMCs (for Reference)

Parameter	Specification Requirement	Remarks
Photo Area		
Acids, Total	< 5 ppbv	Photo Area
Amines, Total	< 4 ppbv	Photo Area
Trimethyl silane (TMS)	< 0.1 ppbv	Photo Area
Condensable	< 50 ppbv	Photo Area
Refractories	< 0.1 ppbv	Photo Area

Ballroom Area and other Areas		
Acids, Total	< 10 ppbv	All Other Areas except Photo
Amines, Total	< 1 ppbv	Reticle Area
Amines, Total	< 30 ppbv	All Other Areas except Reticle and Photo
Ozone (O3)	< 5 ppbv	SOD Process and Diffusion Process Areas
Hydrogen Sulfide (H2S)	< 5 ppbv	Copper Area only
IPA	<10 ppbv	All Other Areas except Photo

8.0 Probe Cleanroom Design:

8.1 Probe Cleanroom Airflow Recirculation

The Recirculation Air Change rate is used to achieve the Cleanliness specification and to achieve the Cleanroom Temperature and Humidity specification. Providing sufficient Air Changes is essential to capture and remove the particles generated in the space, and also to provide cooling and remove the heat load from the space.

The Air Change rate in the Probe Cleanroom space is a function of the Probe Cleanroom volume (Cleanroom Area x Ceiling Height), Ceiling FFU Coverage, and FFU air velocity.

The minimum design requirement and recommended design for FFU Coverage and Air Change Rate is shown in **Table 8**:

Table 8: Probe Cleanroom Environment – Cleanroom Class, Minimum FFU Coverage and Air Change Rate

Space	ISO Class	Minimum FFU Coverage	Minimum Air Change Rate
Probe Cleanroom	ISO Class 6 (Minimum)	Minimum:≥45% ~ 50% Recommended: 50%	Minimum:>120 ~ 140ACH Recommended: 140ACH

Note: Table 8 is calculated based on a 5.0m Cleanroom Ceiling height. For a given FFU coverage and FFU airflow amount, a lower Cleanroom ceiling height will provide a higher Air Change Rate. However, for Cleanrooms with lower ceiling heights, the higher Air Change Rate may be preferred to push the heat generated by tools downwards towards the under raised floor.

8.1.1 The minimum design Air Changes shall be 120 ACH, but the recommended design Air Changes is greater than 140 ACH.

8.1.2 The minimum design FFU coverage shall be 45%, but the recommended FFU coverage is greater than 50%.

8.1.3 To determine the Air Change Rate, the FFU airflow amount shall be calculated based at the FFU air velocity of 0.40 m/s (80 FPM). The FFU selection shall have capability of operating at a minimum of 0.45 m/s (90 FPM), and therefore that the FFU speed can be increased in the future to increase the Air Change Rates.

8.2 Probe Cleanroom Airflow Recirculation through Return Air Chase and Interstitial Plenum

The sizing and design of the vertical Return Air Chase and Interstitial Plenum is important to minimize the pressure losses and enable proper Cleanroom Airflow Recirculation.

This Standard covers the airflow design requirements of the Cleanroom Return Air Chase and Interstitial Plenum, but will not cover the design and construction of Return Air Chase and Interstitial Plenum including building construction, performance and load ratings, structural supports, materials, coatings, constructability, etc.

8.2.1 The velocity in Return Air Chase shall be kept below 2.5 to 3.0 m/s (500 to 600 FPM). The recommended design velocity is 2.0 to 2.5 m/s (400 to 500 FPM). If these prescriptive conditions cannot be fulfilled, relevant engineering calculation is needed to justify that the pressure drop associated with higher velocity is still within acceptable range per case-by-case basis.

8.2.2 The velocity in Interstitial Plenum shall be kept below 2.5 m/s (500 FPM). The recommended design velocity is 2.0 m/s (400 FPM). If these prescriptive conditions cannot be fulfilled, relevant engineering calculation is needed to justify that the pressure drop associated with high velocity is still within acceptable range per case-by-case basis.

8.2.3 There shall be no air infiltration or leakage from the outside spaces into the Return Air Chase or Interstitial Plenum. This includes, but not limited to, the following:

- Building construction seams, joints, and gaps shall be covered with separation panels and sealed to prevent infiltration and leakage.
- All joints in walls, floors, and ceilings shall be of vapor-tight construction.
- Sealing and caulking with cleanroom approved sealant shall be used to seal all gaps between panels and gaps in cut-outs.

Note: The Interstitial Plenum is typically at negative pressure with relation to the Probe Cleanroom and outdoor environment. Therefore, the building structural walls and roof provide the Cleanroom containment area. Negative pressure plenums ensure all air goes through the FFUs and filtration system. However, this introduces the risk of contamination because the building structure is used as walls of the plenum and return air chase. The joints of roof and walls must be sealed vapor-tight to avoid leakage of unconditioned and unfiltered outdoor air from outside into the Interstitial Plenum or Return Air Chase.

8.3 Probe Cleanroom Airflow Recirculation through Under Raised Floor

Sufficient floor openings in the Cleanroom Raised Floor is important to minimize to allow proper airflow path and enable proper Cleanroom Airflow Recirculation.

This Standard covers the airflow design requirements of the Cleanroom Raised Floor tile, but will not cover the design, selection, and construction of the Cleanroom Raised Floor System including system type, performance and load ratings, structural supports, materials, coatings, constructability, etc.

8.3.1 It is recommended to have minimum 50% or more of the Cleanroom floor tile as Perforated Panel type or Grating Panel type, to allow proper Cleanroom Airflow to pass through from Cleanroom Level to under raised floor space. The floor opening coverage % should be based on the FFU coverage.

8.3.2 As Cleanroom tools are being installed and/or during tool move-in, the perforated floor tiles shall be relocated from under the tools to in-between the tool aisles, to maintain the cleanroom recirculation air flowpath. It shall always be confirmed that there is sufficient return air path for the cleanroom recirculation air.

8.3.3 When selecting Cleanroom Raised Floor tiles, the following parameters shall be met:

- Allowable pressure drop shall be between 5 to 10 Pa. The recommend pressure drop is less than 5 Pa.
- The typical airflow rating per floor tile size 600mmm x 600mm (24in x 24in) shall be ~ 500 CMH to 860 CMH each.
- Free area opening for Perforated Panel type: 17-22% opening
- Free area opening for Grating Panel type: 50% opening

8.3.4 In areas with high heat loads and in any other areas where possible, it is recommended to use Grating Panel type floor tile, which has 50% free area opening to allow more Cleanroom Airflow to pass through.

8.3.5 The typical clearance height under the raised floor shall be designed between 1.5 to 1.7 m (5 to 5.5 feet) to provide a sufficient cross-sectional area for Cleanroom air to recirculate back into Return Air Chase and to accommodate for the utility services to support the production tools.

8.3.6 The air velocity under the raised floor shall be checked, and relevant engineering calculation is needed to justify that the pressure drop will be within the acceptable range per case-by-case basis.

8.4 Probe Cleanroom Heat Load

The Cleanroom HVAC System shall be designed to provide sufficient cooling and airflow provisions, to satisfy the Cleanroom Heat Load and maintain the Cleanroom Dry Bulb Temperature within specifications.

8.4.1 The default Probe Cleanroom Design Heat Load is shown in **Table 9,** and recommended to be 1400 W/m2.

Table 9: Design Heat Load for Probe Cleanroom Space

Parameter	Design Requirement	Applicable Spaces / Remarks
Design Heat Load	Minimum: ≥1000~1400 W/m2 Recommended: 1400 W/m2	Probe Cleanroom

8.4.2 During the Project Design Phase, the actual heat load shall be calculated and verified based on the actual Space layout (lighting, FFUs, etc.) tool layout and tool types, and heat output of all tools and equipment.

Note: For Cleanroom spaces with higher heat load based on the actual tool layout and density, the specified Cleanroom design Heat Load shall be higher to meet the actual heat load requirements.

If these Cleanroom spaces also require a higher Air Change Rate based on the actual tool layout and density, the specified design Cleanroom Air Change Rate shall be higher to meet the actual heat load requirements.

For Probe Cleanroom spaces with lower heat load based on actual tool layout and density, the specified Cleanroom Design Heat Load of 1400 W/m2 shall be used. This allows for sufficient cooling capacity for future toolsets and scenarios.

8.4.3 **Note:** Certain equipment inside the cleanroom may generate high amounts of heat and can cause localized temperature fluctuations as a result of effects of radiant heat. It is recommended to locate the Cleanroom Temperature and Humidity sensors away from these tools.

8.5 Cleanroom Pressurization

To prevent or minimize particle migration into the Cleanroom from surrounding areas, the Cleanroom shall be kept positively pressurized relative to the surrounding areas. This positive pressure differential shall ensure that air leakage and particle migration will occur outwards between the Level 3 Cleanroom enclosure and the dirtier surrounding areas.

8.5.1 The Probe Cleanroom Pressure shall be a minimum of 7-8 Pa. The recommended Cleanroom Pressure is 10-12 Pa. Refer to **Table 5** above.

Note: Traditionally, a pressure differential of 12.5 Pa (0.05” w.c.) has been used for Cleanroom design. Industry reference recommends a minimum space pressure of 10 Pa, to minimize particle migration that may potentially occur from opening and closing of Cleanroom doors. However, for Cleanroom operation with door-closed conditions, industry reference recommends a minimum space pressure of 5 Pa. It states that 5pa pressure levels have proven to be sufficient to minimize the particle migration, and higher values of space pressurization will not result in any extra reduction of Cleanroom particle levels.

Some Micron FABs have been able to operate at Cleanroom pressure levels of ~5 to 6 Pa, with no known issues or problems to meet Cleanroom Classification levels, particle count limits, or wafer particle defects.

8.5.2 This Cleanroom Pressure levels shall be measured relative to outside ambient pressure, via the Zero Reference Pipe (ZRP). Differential pressure sensors shall be used to measure the difference between Cleanroom pressure and outdoor air, via the ZRP. The ZRP provides a single common "zero" point that all areas can reference.

The ZRP piping shall be leak-tight stainless steel piping. The ZRP pressure element shall be installed in a stable location outdoors, and protected from potential effects from weather elements including wind, rain, etc.

8.5.3 For Pressure Control, the Direct Pressure-Differential Control (DP) method shall be used, in which a pressure differential sensor measures the pressure difference between the Cleanroom and adjacent space (outside ambient, via the ZRP). Then, the Makeup Air Unit fan speeds shall be automatically modulated, to adjust the Makeup airflow amount supplied to the Cleanroom, to achieve the required pressure differential.

8.5.4 The typical pressure difference level for pressure difference (i.e. cascade) between segregated zones shall be 3 to 5 Pa.

8.6 Cleanroom Dry Cooling Coils

The Cleanroom Dry Cooling Coils (DCC) provide cooling for Cleanroom Temperature control. The Cleanroom Dry Cooling Coils shall be located in Interstitial Plenum in vertical configuration, between the Return Air Chase and Interstitial Plenum Return to the Ceiling FFUs.

8.6.1 The Cleanroom Dry Coil shall be sized, selected, and specified in accordance with **Table 10**.

8.6.2 Medium temperature Chilled Water (MTCHW) shall be provided to the DCC, at supply temperature 12°C or 13°C. The Chilled Water supply temperature shall always be higher than the Cleanroom Dewpoint temperature, to prevent condensation from the piping or coils.

8.6.3 Temperature transmitters shall be provided and installed at the discharge side of each set of Dry Cooling Coils, to monitor the DCC leaving air temperature as it is supplied to Cleanroom.

8.6.4 High Sensitivity Smoke Detector (HSSD) smoke detection shall be provided and installed at the return side upstream of the Dry Cooling Coils, for early smoke detection.

8.6.5 All space around the Cleanroom Dry Coils shall be blanked off and sealed, to ensure that all cleanroom recirculation airflow will route through the Dry Coils. There shall be no bypass air around the Dry Coils that does not pass through the coils.

8.6.6 Piping connections to each Dry Coil unit shall have pressure gauges on supply and return, temperature gauges on supply and return, flexible connections at supply and return, and strainer on the supply connection.

8.6.7 Dry Cooling Coils shall not be installed above the Cleanroom Ceiling directly in the FAB Cleanroom Space, or in the horizontal plane of the Cleanroom Ceiling in the FAB Cleanroom Space. This configuration would create a risk of possible water leakage or condensation, which would result in interruption or damage to Cleanroom tools, equipment, and AMHS.

8.6.8 The Water piping installed directly above the cleanroom ceiling shall be minimized, as much as possible.

Table 10: Dry Cooling Coil Specification for Sizing and Selection

Dry Cooling Coil Specification
Cleanroom Dry Cooling Coil (DCC) shall be selected and specified as follows: 1. The Cleanroom Dry Coil shall be selected based on the following parameters: - Entering Air Temperature: 24.0°C DB / 15.3°C WB (9.2°C DP) - Leaving Air Temperature: ≤ 20.0°C DB - Entering Water Temperature: 12 or 13°C - Leaving Water Temperature: 19 or 20°C 2. The velocity in Cleanroom Dry Coil shall be kept below the maximum of 2.5 m/s (500 FPM). The recommended design velocity is 2.0 m/s (400 FPM). 3. The pressure drop through the Cleanroom Dry Coil shall be kept below the maximum of 50 Pa (0.20” w.g.). The recommended pressure drop is less than 40 Pa (0.16” w.g.) 4. Cooling coil sizing and capacity shall be determined according to the required air volume and cooling capacity. It is recommended to use coils with minimum 2-3 rows to achieve the required cooling capacity. 5. Coil shall be drainable with threaded plugs at the headers to allow venting and draining. Provide the drains and vent taps. 6. Coil header shall have inlet connections at the bottom of supply headers at the “air-off” or “air-on” face of coils (as specified) and outlet connections at the top of return headers at the “air-on” face of coils. 7. Air Leakage: a. Minimize air leakage through gaps between the fin ends and frames. b. Use baffle plates to close the space between the coil frame and the surrounding structure or equipment. c. Seal gaps between coils and surrounding structure or equipment using non-hardening mastic. 8. Pre-completion Production Tests: a. Factory shall perform Pneumatic leak test; Pneumatically test coils for leakage by submerging in

Dry Cooling Coil Specification
warm water and applying the test pressure, or by static water pressure for at least 1 hour. Certification test certificates from coil manufacturer required for each coil. b. Perform full system test at minimum test pressure of 10 Bar. 9. Installation a. During delivery, assembly, and installation, protect coils so fins and flanges are not damaged. Replace loose and damaged fins. Comb out bent fins. b. Install coils so that fluid and air flow directions are counter flow. c. Support coils using prefabricated aluminum alloy or stainless steel frames **Acceptable Manufacturers –** Aerofin, Johnson Controls, Huntair, Heatcraft, Trane, Precision Coils, or approved equal

8.7 Cleanroom Fan Filter Units

Fan Filter Units (FFU) are used to provide final filtration of Cleanroom Recirculation air and then to supply the air to the Cleanroom FAB Level. FFUs shall draw in air from the interstitial plenum and return air shaft, filter the air through the ULPA filters, and then supply air into cleanroom at constant airflow rate.

Fan Filter Units provide high reliability, low first cost, and low operating cost. If one FFU fails, only a very small percentage of the overall air volume to the Cleanroom will be lost. The failed unit can then be easily identified by the FFU Control system for troubleshooting and replacement.

8.7.1 The Fan Filter Unit shall be sized, selected, and specified in accordance with **Table 11**.

8.7.2 For the Cleanroom Air Change Rate calculation purposes, the speed of 0.40 m/s (80 FPM) shall be used.

8.7.3 The FFU shall be capable to accommodate a future frame above the FFU air inlet opening, to add an AMC filter.

8.7.4 The FFUs shall be arranged across the Cleanroom ceiling to provide uniform airflow throughout the entire Cleanroom. For example, 50% FFU coverage would be arranged in a “diamond X-grid” pattern so that every other ceiling tile contains an FFU.

8.7.5 During initial construction and initial start-up, a temporary blanket type construction pre-filter shall be used to cover the FFU inlet air opening. This will protect and lengthen the lifetime of the FFU ULPA filter. After construction is complete, the temporary pre-filter shall be removed.

Table 11: Fan Filter Unit Specification for Sizing and Selection

Fan Filter Unit Specification
The Fan Filter Unit size shall match the Cleanroom Ceiling size. Nominal FFU size is 1200mm x 1200mm (48” x 48”). Fan Filter Unit (FFU) shall be selected and specified as follows: 1. Provide fully factory-assembled, packaged, pre-wired FFU units consisting of: a) intake opening

Fan Filter Unit Specification
b) direct-drive fan with high performance impeller, backwards curved blades, dynamically balanced c) DC Brushless motor (also known as Electronically Communicated (EC) Motor) d) addressable controller with feedback e) control software f) sound attenuation liner g) ULPA filter 2. The Fan Filter Units shall be selected with minimum operation range capability of 0.20 m/s to 0.45 m/s (90 FPM). It is recommended to select with operation capability of up to 0.55 m/s for higher performance. 3. The ULPA filter shall be rated U15, 99.9995% efficient for Most Penetrating Particle Size (MPPS) of 0.12 microns size. 4. The initial pressure drop rating for ULPA Filter shall be less than 120 to 170 Pa @ 0.45 m/s. It is recommended to be less than 130 Pa @ 0.45 m/s. 5. The FFU shall have minimum fan static pressure rating of 200 Pa. The recommended fan static pressure rating is greater than 250 Pa. 6. FFU shall be FM approved. 7. FFU shall be compatible with the Cleanroom ceiling grid system specified. FFU shall fit into ceiling grid geometry without any gaps or leakage. 8. Units shall be designed with sufficient structural strength to be capable of being supported at each corner. Units shall be walkable and support loading on top of unit. 9. FFU shall be manufactured in a clean environment. 10. Fan: a) Fan shall be high efficiency centrifugal plug fan with EC motor, variable flow capacity, direct drive with backwards curved centrifugal impeller, sealed bearings, external rotor motor rated for continuous duty, furnished with internally wired thermal overload protection. Bearing lifetime shall be 80,000 hours. 11. FFU Control System shall be provided with fully integrated software package, tested in service, for monitoring and control of FFU operation, which includes: 1. Unit starting and stopping

Fan Filter Unit Specification
2. Adjusting speed set point 3. FFU status monitoring and alarms 4. FFU runtime counter a) In the event of FFU remote monitoring network or power failure, configure controls so unit operates at last condition b) Each FFU shall be able to operate individually by a central control and monitoring system c) FFU shall have indicator lights to indicate status of malfunction. 12.Motors: a) Provide high efficiency electronically commutative (EC) brushless DC type motors with built-in power factor controller. b) Provide motors of protection class IP44, motor insulation class A. 13. Filter: a) Provide Ultra Low Particulate Arrest (ULPA) filter, rated U15, with minimum DOP efficiency of 99.9995% rated on 0.12 microns size. b) ULPA Filter material shall be PTFE media c) The filter media shall be FM approved. d) Provide FFU with gasket seal between the filter and the FFU housing e) Filters shall be factory tested to demonstrate compliance with efficiency, leak, and pressure drop specifications. Provide all filters with factory test certificate of compliance 14. Provide FFU with the following Labels and Listings: a) UL 507 / CAN CSA C22.2 no. 113-M 1984 b) UL 2111 c) UL 1917 / CAN CSA C22.2 d) All electrical components shall be UL listed and labeled. **FFU Approved Manufacturers:** M+W Products (Exyte), Nortek Air Solutions, Gebhardt, AAF Flanders, or approved equal **ULPA Filter Acceptable Manufacturers:** AAF, Camfil, Trox, or approved equal

8.8 Cleanroom Makeup Air

Makeup air is used to provide Cleanroom Humidity (i.e. Dewpoint) control as well as maintain positive pressurization inside the Cleanroom to prevent particle contamination and air infiltration. Makeup air shall be provided to compensate for the exhaust and cleanroom leakage.

This Standard covers the Makeup Airflow design requirements corresponding to proper Cleanroom Design. However, this Standard will not cover the design, selection, and construction of the Cleanroom Makeup Air Units including equipment and system type, performance ratings, materials, coatings, constructability, etc. Refer to the Cleanroom Makeup Air Design and Construction Standard for this information.

The minimum Makeup Air provided shall be calculated per **Table 12**:

Table 12: Guideline for Cleanroom Makeup Air Requirement Calculation

- Cleanroom Makeup Air Requirement = Total Exhaust inside Cleanroom + Total Pressurization/ Leakage
- Total Exhaust inside Cleanroom = Tool Exhaust (from all levels) + Exhausts inside Cleanroom + Space/Area Exhaust inside Cleanroom
- Total for Pressurization/Leakage = 1 Cleanroom Air Change per hour (based on Cleanroom Volume x 1)

8.8.1 Create a Room Pressure and Flow (P&F) diagram for the Cleanroom, to show the following:

a) Airflow design settings (values) of all supply, return, and exhaust amounts
b) Desired room pressure value and tolerance
c) Leakage flow directions due to room pressure differentials

The Cleanroom Air Balance calculation shall be provided to show the amount of Makeup Air required versus Exhaust Air, Cleanroom Leakage, and Pressurization at Full Build conditions.

8.8.2 Makeup air shall be ducted to the Return Air Chase, and the MAU supply shall be discharged directly to the middle of the Return Air Chase.

8.8.3 Sufficient mixing shall be ensured between makeup air and recirculation air to prevent temperature and humidity stratification and imbalance.

8.8.4 Makeup air supply shall have capability to be provided at following parameters:

- Normal Makeup Air Supply Design Setpoint shall be Dry Bulb 22.5°C, Dew Point 9.0°C
    - The Makeup Air Supply Dry Bulb Temperature shall have capability to provide as low as 15°C, and capability to provide as high as 23°C.
    - The Makeup Air Supply Dew Point Temperature shall have capability to provide as low as 8.3°C and capability to provide as high as 10.3°C.

  During winter seasons, it may be required to supply the Makeup Air discharge at a higher Dewpoint temperature than the Cleanroom Dewpoint temperature, to provide additional humidification:

    - Dew Point Temperature: Makeup Air Supply shall have capability to provide as high as 10.3°C for humidification.

  During summer seasons, it may be required to supply the Makeup Air discharge at a lower Dewpoint temperature than the Cleanroom Dewpoint Temperature, to provide additional de-humidification:

    - Dew Point Temperature: Makeup Air Supply shall have capability to provide as low as 8.3°C for dehumidification.

8.8.5 For sites with relatively constant climate year-round (i.e. Singapore, Malaysia), either fixed control setpoint or Auto-Cascade control setpoint can be considered for the Makeup Air discharge dewpoint controls.

a. The system shall have capability to operate in either fixed control setpoint or Auto-Cascade control setpoint. The method of control can be selectable by the Operations team.

8.8.6 For sites with seasonal climate changes for summer, winter, fall, and spring, (i.e. USA, Japan, Taiwan, China), it is recommended to use Auto-Cascade control capability to automatically adjust the Makeup Air Discharge Dewpoint setpoint.

a. Upper and Lower limit clamps shall be set to prevent Makeup Air Discharge Dewpoint from drifting excessively from the normal setpoint.

b. Upper and Lower alarm limits shall be set to notify Operations if the Makeup Air Discharge Dewpoint drifts excessively from the normal setpoint.

c. The system shall have capability to operate in either fixed control setpoint or Auto-Cascade control setpoint. The method of control can be selectable by the Operations team.

8.8.7 The Makeup air unit shall have the following filters and filter ratings, at a minimum:

a) Pre-filter rated at 25 to 35% (G4, MERV 7)
b) Intermediate filter rated at 85-95% (F7-F8, MERV 13-14)
c) 2 Racks of AMC filters for Ozone, Acids, Bases, as needed based on the site's ambient air conditions and upstream AMC challenge
d) Final HEPA filter rated at 99.95% (HEPA H13)

Note: Makeup air units with air washers shall have capability to remove SO2 and NH3 via the air washer, to meet the minimum requirements below:

AMC	Inlet concentration range	Removal Efficiency
NH3	30ppb <Inlet conc. <300ppb	≥ 90% or ≤ 3ppb
SO2	40ppb <Inlet conc. <400ppb	≥ 90% or ≤ 3ppb

8.8.8 Makeup air intakes shall not be placed immediately downstream under prevailing winds of exhaust discharge airstreams and cooling tower discharge airstreams, the former there is potential risk of cross-contamination of exhaust, the later, there is potential risk of warm and humid air draw into the Makeup air intakes. CFD Modeling shall be performed to confirm that there is no cross contamination of exhaust into the Makeup air intakes.

8.8.9 There are observations that the Probe Cleanroom relative humidity (RH%) tends to be at lower level and occasionally out of the lower specified RH% limit. The possible root causes may be summarized as follows:

a) Occasionally, CDA with low humidity content is being directly dispersed from the Probe tools into the Cleanroom environment during its operation (e.g., internal purging).
b) Probe tools are typically requiring lower flowrate of process exhaust due to its nature of operation. This might be a contributing factor to the accumulation of CDA inside the Cleanroom environment.
c) Due to the lower flowrate of exhaust air, CRMAU make-up air entering the Cleanroom will mostly be at limited flowrate. As the make-up air flow is the main mechanism in regulating the Cleanroom RH%, its effectiveness is adversely affected.

Thus, it is advisable to have a proper assessment on this potential issue and make appropriate provision to mitigate the impacts during the Cleanroom operation.

8.9 **Cleanroom CFD Modelling** – During Design phase, the Cleanroom conditions shall be checked using Computational Fluid Dynamics (CFD) modelling software. CFD modelling shall be used to simulate and determine the following Cleanroom conditions, and make the necessary improvements to the Design:

a) Airflow pattern
b) Temperature distribution
c) Particle concentration
d) Relative humidity distribution
e) Impact of room pressurization
f) As part of the CFD Modelling and evaluation, the CFD model shall include the AMHS Track, AMHS storage and maintenance walkways, FAB level exhaust ductwork, and other potential obstructions below the Cleanroom Ceiling
g) CFD Modelling shall also be performed on the exterior building Makeup Air and Exhaust systems, to confirm that there is no cross-contamination of Exhaust streams into the Makeup air intakes.

8.10 Cleanroom Temperature Control Zoning

The Cleanroom shall be separated into temperature control “zones” based on the location of Dry Cooling Coils and their corresponding Return Air Chase. Feedback from the Cleanroom Temperature and Humidity (T&H) sensors shall control a single Dry Coil unit or group of

Dry Coil units. Refer to **Figure 2** for example of Cleanroom Temperature Control Zoning.

8.10.1 Approximate Cleanroom Temperature Control Zone size shall be 500 to 750m2 each.

8.10.2 The zone shall be controlled by the Dry Cooling Coils facing that zone.

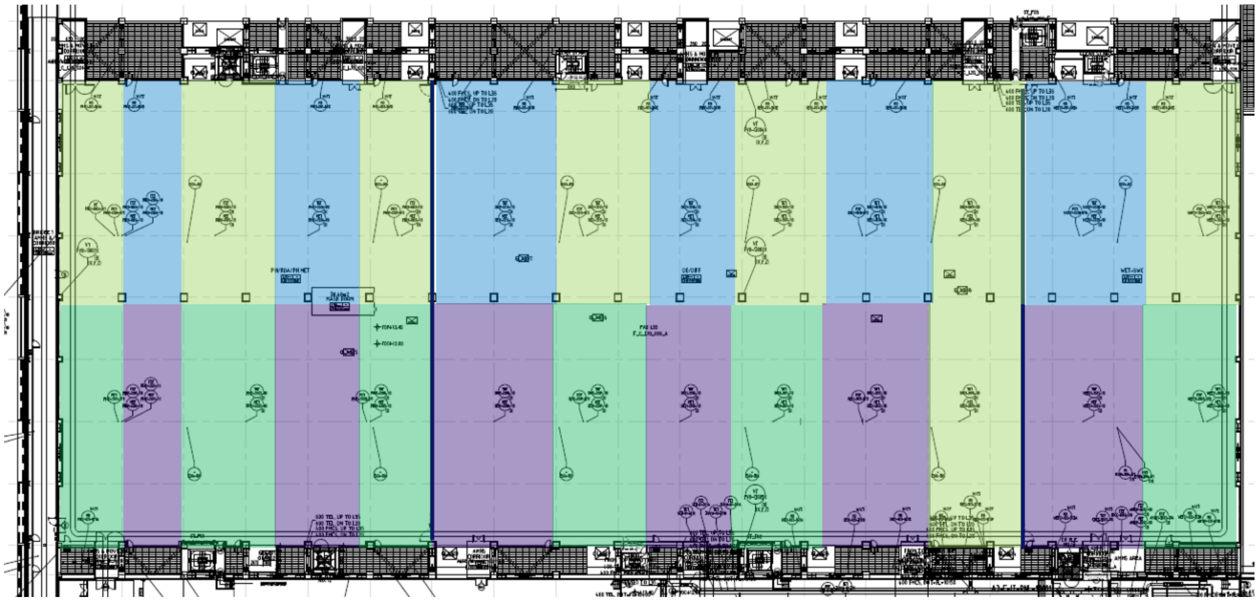
8.10.3 Distance in between the Return Air Chases shall be limited to be below 60m. It is recommended to limit the distance between Return Air Chase to any point in the Cleanroom to be below 30m, for improved airflow performance.

8.10.4 Each Cleanroom zone shall have 2 T&H sensors. The sensors and their respective probes shall be located in a manner that the temperature readings are similar; it is recommended to locate the 2 T&H sensors side-by-side. This is to provide better temperature control via averaging function.

8.10.5 Avoid stagnant areas throughout the Cleanroom, in which there would be no supply of conditioned air from Dry Cooling Coils. The design and location of Return Air Chases and Dry Cooling Coils shall be configured to provide uniform and even distribution of conditioned supply air throughout the entire Cleanroom.

8.10.6 There shall be one control valve station for each set of Dry Cooling Coils, associated with a Temperature Control Zone. The Control Valve station shall have isolation valves, and a smaller size bypass with isolation valve around the control valve.

Figure 2 – Example Cleanroom Temperature Control Zoning for FAB Ballroom



8.11 Cleanroom Temperature & Humidity Sensors

Ceiling mounted Temperature and Humidity Sensors shall be used in each Temperature Control Zone.

8.11.1 T&H Sensor shall be located at directly below the FFU discharge, at a height of 100mm below ceiling height, and at a distance 200mm into the face of the FFU. Refer to **Figures 4 and 5** for example.

This sensor location allows the sensor to sense adequate mixing of the FFU supply air (from the DCC and interstitial plenum) mixed with the Cleanroom Heat sources.

Regarding temperature sensor location, Industry reference recommends that it is more important to have precise control rather than accurate control, and to provide for repeatability of results. Therefore, the temperature sensors are typically located in plenum area (measuring coil discharge temperature) or just below the Cleanroom ceiling, to measure the temperature of the supply air.

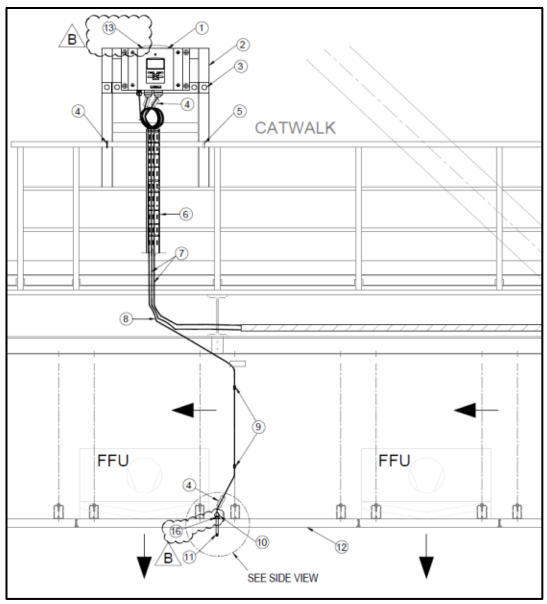
a. Sensor shall be located away from AMHS track.

b. Sensor shall be located away from other obstructions and heat sources, such as exhaust ductwork, utility piping, etc.

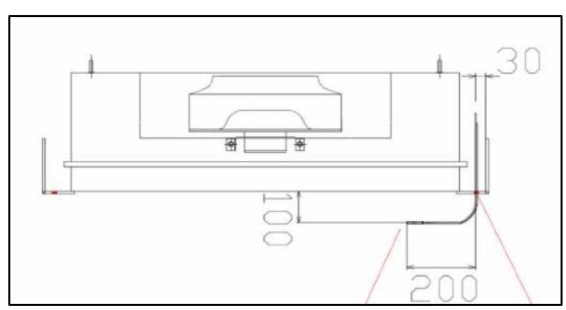
8.11.2 Transmitter module shall be mounted up at Interstitial Plenum along catwalk or grating, to allow easy access for calibration and troubleshooting. Refer to **Figure 3** for example of Transmitter mounting detail.

8.11.3 The probe shall have excess cable length of minimum 10m, to allow for future relocation of the probe.

Figure 3 – Example Cleanroom T&H Transmitter and Probe Mounting Detail



Figures 4 & 5 – Recommended Cleanroom T&H Probe Mounting Location



8.12 Cleanroom Building Envelope and Cleanroom Ceiling

This Standard will not cover the design, selection, and construction of the Cleanroom Building Envelope and/or Ceiling System including system type, performance and load ratings, structural supports, materials, coatings, constructability, etc.

The Cleanroom envelope shall be sealed to prevent any infiltration leakage from outside space, or exfiltration leakage to outside space.

It is critical to ensure the Return Air Plenum and Plenum is sealed, as these spaces may be at negative pressure relative to outside. If the spaces are not sealed properly, they may cause outside air to infiltrate and affect Cleanroom Temperature and Humidity control, AMC levels, or load up the FFU ULPA filters with particulate.

Cleanroom ceiling considerations to be provided include, but is not limited to, the following:

a) Ceiling shall have proper gasketing to ensure it is leak-tight and air-tight when pressurized.
b) FFUs shall sit directly in the Cleanroom Ceiling.
c) Cleanroom Ceiling shall have blank walkable access panels in locations without FFUs.
d) Cleanroom Ceiling shall have suitable penetrations for components and devices. These include, but are not limited to, temperature, humidity, and pressure sensors, sprinkler heads, lighting, I&C, telecom, and other life safety devices.
e) Sheet metal flashing, closures, gaskets and sealing at columns, perimeter walls, and penetrations through blank panels.
f) Changeable layout and locations for filters, blank panels, and luminaries.
g) The space above the Cleanroom ceiling shall have suitable tie-off fall protection, that is OSHA compliant.

For Cleanroom Segregated areas, the entire Interstitial Plenum area and Subfab area of the FAB shall be divided into different areas by means of wall separations, floor separations, plenum separations, and ceiling separations. These separations shall allow for dedicated individual temperature, humidity and pressure zoning. Ceiling separator shall be installed from ceiling grid framework up to the ceiling purlins. The plenum separators shall be installed from the ceiling purlins up to the plenum cap. Maintenance Access Panels should be provided to access these enclosed areas.

8.13 Probe Gowning Rooms Cleanroom Design

The Cleanroom HVAC System for the Probe Gowning Rooms shall be designed to the specifications as shown in **Tables 17 and 18.**

Table 17: Probe Gowning Room Cleanroom Specifications

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	ISO 14644-1 Class 7	Probe Gowning Room
Cleanroom Temperature	22.5 ± 2 °C [72.5 ± 3.5 °F]	Probe Gowning Room
Cleanroom Humidity	42 ± 7* %RH	Probe Gowning Room *due to Non-Critical for RH Spec
Cleanroom Pressurization	Minimum: 2 Pa Recommended: 3-5 Pa	Probe Gowning Room

Table 18: Probe Gowning Room Cleanroom Design

Space	ISO Class	Heat Load	Minimum FFU Coverage	Minimum Air Change Rate
Probe Gowning Room	ISO Class 7	Recommended 250 W/m2	Minimum: >28% Recommended: >28%	Minimum: >60 ACH Recommended: >60 ACH

a) Gowning Area is typically single level Cleanroom only and does not have a Subfab Level or Interstitial Level. The cleanroom airflow shall be supplied from the Ceiling mounted FFUs, down to sidewall return grills at floor level, with Dry Cooling Coils mounted behind the sidewall return grille. The recirculation airflow shall then return up the sidewall vertical return air chase, to above the ceiling plenum, and back to the FFUs.

b) Gowning Area shall have its own dedicated Cleanroom Temperature & Humidity sensors, separate from the Main Ballroom Area.

c) Gowning Area shall have its own dedicated Cleanroom Pressure sensors, separate from the Main Ballroom Area.

d) Gowning Area pressurization shall be cascaded with lower pressure in comparison to the Ballroom Area. The recommended Cleanroom Pressure in Gowning Area shall be 3 to 5 Pa.

8.14 AMHS Link Bridge Cleanroom Design

The Cleanroom HVAC System for AMHS Link Bridge shall be designed to the specifications as shown in **Tables 19 and 20.**

Table 19: AMHS Link Bridge Cleanroom Specifications

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	ISO 14644-1 Class 6	AMHS Clean Link Bridge
Cleanroom Temperature	22.5 ± 2 °C [72.5 ± 3.6 °F]	AMHS Clean Link Bridge
Cleanroom Humidity	42 ± 7* %RH	AMHS Clean Link Bridge *Due to Non Critical for RH Spec

Table 20: AMHS Clean Link Bridge Cleanroom Design

Space	ISO Class	Heat Load	Minimum FFU Coverage	Minimum Air Change Rate
AMHS Link Bridge	ISO Class 6	Recommended 250 W/m2	Minimum: >28% Recommended: >28%	Minimum: > 60 ACH Recommended: >60 ACH

a) AMHS Link Bridge is typically single level Cleanroom only and does not have a Subfab Level or Interstitial Level. The cleanroom airflow shall be supplied from the Ceiling mounted FFUs, down to sidewall return grills at floor level, with Dry Cooling Coils mounted behind the sidewall return grille. The recirculation airflow shall then return up the sidewall vertical return air chase, to above the ceiling plenum, and back to the FFUs.

b) AMHS Link Bridge shall have its own dedicated Cleanroom Temperature & Humidity sensors, separate from the Main Ballroom Area. These shall be used for temperature control of the dry cooling coils.

c) AMHS Link Bridge shall have its own dedicated Makeup air branch ductwork supplying to the AMHS Link Bridge. This branch ductwork shall have manual volume dampers for balancing. The Makeup air shall discharge into the plenum level above the FFUs.

d) The AMHS Link Bridge shall be sealed to prevent any infiltration leakage from outside space, or exfiltration leakage to outside space.

8.15 Probe Decon Room Cleanroom Design

The Cleanroom HVAC System for Probe Decon Rooms shall be designed to the specifications as shown in **Tables 21 and 22.**

Table 21: Probe Decon Room Cleanroom Specifications

Parameter	Specification Requirement	Applicable Sites / Remarks
Cleanroom Particles	ISO 14644-1 Class 7	Probe Decon Cleanroom
Cleanroom Temperature	22.5 ± 2 °C [72.5 ± 3.6 °F]	Probe Decon Cleanroom
Cleanroom Humidity	42 ± 7* %RH	Probe Decon Cleanroom *Due to Non Critical for RH Spec

Table 22: Probe Decon Rooms Cleanroom Design

Space	ISO Class	Heat Load	Minimum FFU Coverage	Minimum Air Change Rate
Probe Decon Room	ISO Class 6	Recommended 250 W/m2	Minimum: >28% Recommended: >28%	Minimum: > 60 ACH Recommended: > 60 ACH

a) Probe Decon Cleanroom is typically single level Cleanroom only and does not have a Subfab Level or Interstitial Level. The cleanroom airflow shall be supplied from the Ceiling mounted FFUs, down to sidewall return grills at floor level, with Dry Cooling Coils mounted behind the sidewall return grille. The recirculation airflow shall then return up the sidewall vertical return air chase, to above the ceiling plenum, and back to the FFUs.

b) FAB and Probe Decon Cleanroom shall have its own dedicated Cleanroom Temperature & Humidity sensors, separate from the Main Ballroom Area. These shall be used for temperature control of the dry cooling coils.

c) FAB and Probe Decon Cleanroom shall have its own dedicated Makeup air branch ductwork supplying to the FAB Decon Room. This branch ductwork shall have manual volume dampers for balancing. The Makeup air shall discharge into the plenum level above the FFUs.

8.16 **Emergency De-Smoke Exhaust System** – An Emergency De-smoke exhaust system shall not be required, unless specifically required by local code or AHJ for emergency smoke removal purposes.

If an Emergency De-smoke exhaust system is required, the minimum design flowrate of smoke purge shall be 55m3/h/m2 for Ballroom areas, as outlined in NFPA 318-2002 Edition, Annex A, Clause A.5.6.4, or as required by local code and AHJ.

The Emergency De-smoke exhaust system design and specification requirements will not be covered in this Standard.

8.17 **Other General/Miscellaneous:**

a) **Code Requirements -** Cleanroom design shall comply with local code and AHJ requirements based on the level of Hazardous Occupancy Classification.

b) **Equipment Layout –** All HVAC Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment and to allow move-in and move-out of large components during maintenance and repair (such as coils, motors, filters, etc.).

c) **Water Piping –** Water piping shall be minimized directly above Probe Cleanroom and minimized inside the Interstitial Plenum, as much as possible.

d) **Fire Sprinkler -** Probe Cleanroom shall be covered via Wet Pipe type Fire Sprinkler system. The requirements for whether Fire Protection is required in the Interstitial Plenum depends on the local code and AHJ. Refer to the Fire Suppression Design and Construction Standard for additional requirements.

9.0 Design for Sustainability

The following energy efficiency and Sustainable design elements shall be included:

9.1 **Cleanroom Air Recirculation Low Pressure Drop design** – Design and selection of recirculation shall be to minimize air velocity and pressure drop, to minimize FFU fan energy consumption. This includes lower air velocities through the return air chase, interstitial plenum, and across the Dry Cooling Coils.

9.2 **Minimize Cleanroom Leakage and Pressurization** – The amount of cleanroom leakage shall be kept as minimal as possible via tight building construction. This will allow the Cleanroom pressure levels to be kept lower, and to reduce the amount of makeup air required for pressurization and also to compensate for temperature/humidity impact from infiltration.

The lower Cleanroom Pressurization spec of 7-8Pa minimum and 10-12Pa is recommended versus higher levels of Cleanroom Pressurization, to minimize the amount of Makeup Air needed to over-pressurize the cleanroom to the higher Pressurization Spec.

9.3 **Minimize Cleanroom Exhaust and Makeup Air** – The amount of Cleanroom exhaust shall be kept as minimal as possible via proper balancing of production tool exhaust and reducing the exhaust loads wherever possible. This will allow reducing the amount of Cleanroom Makeup air required to compensate for exhaust.

The exhaust quantities used by each production tool shall be balanced to their design flowrates and actual requirements, to minimize the amount of excess exhaust used (and excess resultant Cleanroom Makeup air that would be required).

Another opportunity to reduce exhaust is to identify where General Exhaust can be discharged directly into the Cleanroom under raised floor or Return Air Chase, to minimize the amount of air exhausted to outside (and the resultant Cleanroom Makeup air that would be required). When performing this exhaust recovery, it shall be confirmed that the General Exhaust consists of heat exhaust only, and does not include any corrosive/ toxic/ flammable/ hazardous exhaust fumes.

9.4 **FFU Adjustable Speed Controllers** – The FFUs shall be equipped with EC type motors for higher energy efficiency, and adjustable speed controllers to allow more FFU efficient operation at part speed.

10.0 Design for SMART Facilities

The following SMART Facilities design elements shall be considered:

10.1 **Cleanroom Particle Monitoring**: It is recommended to monitor Cleanroom Particle levels via online continuous monitoring system.

10.2 **Cleanroom AMC Monitoring:** It is recommended to monitor Cleanroom AMC levels via online continuous monitoring system.

10.3 **Makeup Air Unit AMC Monitoring:** It should monitor the AMC levels at the Makeup Air Unit discharge (and possibly the Makeup Air Unit intake or outdoor ambient levels), via online continuous monitoring system.

11.0 Accessories:

11.1 The following accessories which shall be installed: See **Table 23**.

Table 23: Accessories to be installed

Accessories	Type and Installation Location	Remarks
Piping Insulation	• Install for all CHW piping throughout the system	In accordance with Spec Section 15260 (or equivalent Spec)
Ductwork Insulation	• Install for all MAU Supply Air ductwork in unconditioned spaces or utility area spaces	In accordance with Spec Section 15260 (or equivalent Spec)
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)

Accessories	Type and Installation Location	Remarks
Flexible Connections	• Install at all CHW connections to Dry Cooling Coils • Install at all ductwork connections to MAU units and equipment	
Pressure Gauges	• Install at all CHWS/R connections to Dry Cooling Coils	
Temperature Gauges	• Install at all CHWS/R connections to Dry Cooling Coils	
Air Vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto Air Vents at all high point locations for CHW • Install Auto Air Vents at main CHW piping and vertical CHW piping connected to the Dry Cooling Coils	
Drain Ports	• Install at all low point locations • Install at all Dry Cooling Coils	
Valves	• Install valves at all Dry Cooling Coils to allow isolation. • All butterfly valves shall be rated for “bi-directional” duty with positive shutoff in both directions • All butterfly valves shall be lug-style	In accordance with Spec Section 15101 (or equivalent Spec)
Control Valves	• Provide control valve type and quantity as outlined in this standard	
Condensate Drain	• Drain piping shall be provided to collect condensate from the condensate system, and to discharge to the Industrial Waste (IW) system or other approved system.	
Labeling and Identification	• “CHWS” and “CHWR” shall be used for Chilled Water Supply and Return • “MAU” shall be used for Makeup Air Supply Ductwork • Provide identification labels at minimum of every 6m distance and change of direction • Labels for CHWS and CHWR piping shall be Green Background with White Lettering	In accordance with Spec Section 15190 (or equivalent Spec)

12.0 Electrical Design:

12.1 **Power Segregation** – Electrical power supply for Makeup Air Units and FFUs shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer). The quantity of equipment served from each upstream power source shall be split evenly.

a) The Electrical power wiring for FFUs shall be arranged in an “alternate” configuration, so that every other FFU is from a different power source. This way, upon the loss of one power source, every other “alternate” FFU would still continue operation to provide uniform airflow across the Cleanroom.

12.2 **Un-interrupted Power for Controls** – 100% of Cleanroom HVAC system controls including PLCs, Local Control Panels (LCPs), FFU System Controller, FFU Control Power, and other auxiliary control power shall be connected to an Uninterrupted Power source (such as UPS or DC Bank) to allow continuous operation of the Cleanroom HVAC system during momentary power interruption or voltage sags, spikes, and fluctuation events.

12.3 **Emergency Power - MAU** – 50% of MAU units shall be on Emergency power and shall continue to run during a normal power loss. This is to maintain positive pressurization in the cleanroom and compensate for the exhaust that will still be operating during a power loss. Refer to the Makeup Air Unit System Standard for additional details and requirements.

12.4 **Emergency Power - Exhaust** – 50% of Exhaust fans shall be on Emergency power and shall continue to run during a normal power loss. This is per NFPA and IBC code requirements. Refer to the Exhaust System Standard for additional details and requirements.

12.5 **Emergency Power - FFU** – 50% of FFUs shall be on Emergency power and shall continue to run during a normal power loss. This is to maintain minimum cleanroom recirculation airflow and cleanroom classification during a power loss.

a) The FFUs with Cleanroom Temperature & Humidity sensors located beneath shall be connected to Emergency Power. Therefore upon a loss of Normal Power, the T&H sensor would still be able to sense the Cleanroom conditions accurately, to keep the Cleanroom control function ongoing.

12.6 **Emergency Power – Chilled Water Distribution Pumps (either Primary or Secondary Pumps) serving Cleanroom Dry Cooling Coils** – It is recommended to have 25 to 50% of the Chilled Water

Distribution Pumps serving the Dry Cooling Coils (either Primary or Secondary Pumps) to be on Emergency power, to be able to continue to run during a normal power loss. This is to maintain minimum chilled water flow recirculation through the Dry Cooling Coils during a power loss. This will help to minimize the impact to Cleanroom Temperature, and the amount of time it takes to recover the Cleanroom Temperature back within spec, during an extended power loss event.

13.0 Instrumentation:

13.1 The following Instrumentation shall be installed and monitored/ controlled via FMCS. See **Table 24**.

Table 24: Instrumentation to be installed

Instrumentation	Type and Installation Location
Cleanroom Temperature and Humidity Transmitters	• Install 2 at each Temperature Control Zone (Quantity: 1 duty and 1 backup)
Cleanroom Differential Pressure Transmitter	• Install minimum 3 in FAB Cleanroom Ballroom space
Dry Cooling Coil Leaving Air Temperature Transmitter	• Install at each Dry Coil Unit

13.2 The corresponding Cleanroom Dewpoint temperature shall also be shown on the SCADA screen for each Temperature and Humidity sensor.

14.0 Construction Stage Milestones

The following Construction Stage protocol shall be followed, in accordance with the Global Construction Cleanroom Protocol Standard.

a) **Stage 1 –** Rough Construction Stage - Building Shell Enclosed.
b) **Stage 2 –** Establishing the Clean Zone, Clean Zone Perimeter Boundaries Enclosed.
c) **Stage 3 -** Cleanroom Walls, Floors, and Ceiling Assembly Completed. One or more MAU units installed and operational in “Manual” mode, with filters installed including HEPA Filters, for early run operation to achieve start of Cleanroom Blowdown.
d) **Stage 4 –** Initial Cleanroom Certification; As-Built Status Achieved; Ceiling FFUs with ULPA Filters installed and operational.

e) **Stage 5 –** Intermediate and Final Qualification of Cleanroom and Installation of Production Equipment; Operational Status Achieved; Certification Testing.

- Construction pre-filters on FFUs and MAUs shall be removed after the Final Qualification of the Cleanroom and prior to installation of Production Equipment.

- AMC Filters shall be installed on FFUs and MAUs prior to Final Qualification of the Cleanroom and/or move-in of first Production Tool, in the designated area requiring AMC Filters.

15.0 Cleanroom Certification and Testing:

15.1 **Testing Requirements:** The following Cleanroom Certification Tests shall be performed during Cleanroom Certification:

d) Cleanroom Ceiling and Ceiling/FFU Leakage Test
e) FFU air velocity verification
f) Cleanroom Classification via ISO 14644-1:2015 – Airborne Particle Test
g) Lighting Level test – minimum one point per 100m2
h) Sound Pressure Level – minimum one point per 100m2
i) Room Pressurization
j) Temperature and Humidity uniformity – minimum one point per 100m2, at a height of 1.0 to 1.2m above the Cleanroom Raised Floor
k) Floor resistance measurement – minimum one point per 100m2
l) Floor Vibration Testing
m) MAU HEPA Filter Leakage Test

15.2 **Cleanroom Construction Status Definitions:**

a) **As-Built** - Condition where the cleanroom or clean zone is complete with all services connected and functioning but with no equipment, materials, or personnel present (ISO 14644-1:2015).

b) **At-Rest -** Condition where cleanroom or clean zone is complete with equipment installed and operating in a manner agreed upon, but with no personnel present (ISO 14644-1:2015).

c) **Operational -** Agreed condition where the cleanroom or clean zone is functioning in the specified manner, with equipment operating and with the specified number of personnel present (ISO 14644-1:2015).

15.3 **Initial Certification** - Completed once the “As-Built” status is achieved. Testing is to demonstrate that airborne and surface particle counts, temperature and humidity, surface resistance, and air balance and flow are within project specifications. Testing is also to confirm that the ceiling grid and ULPA/HEPA filters are sealed and functioning as designed.

15.4 **Intermediate Certification** - Completed once the "At-Rest" status is achieved. Testing is to demonstrate that airborne and surface particle counts, and temperature and humidity are within project specifications.

15.5 **Final Certification** - Completed once the "Operational" status is achieved. Testing is to demonstrate that all environmental criteria are within project specifications.

16.0 Control Sequence Philosophy:

16.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 25**. Refer to the specific Sequence of Operations for each project for specific details.

Table 25: System Control Logic

Control Parameter	Control Logic
Cleanroom Drybulb Temperature Control	• Cleanroom Dry-bulb Temperature control shall be achieved via the Dry Cooling Coil modulating control valves. • Each control valve position shall modulate based on the Cleanroom Dry-bulb Temperature from the corresponding temperature control zone.
Cleanroom Humidity Control	• Cleanroom humidity control shall be achieved via Dewpoint Control of the MAUs. • The MAU discharge dewpoint shall be achieved via fixed Control or Auto-Cascade Control based on Cleanroom dewpoint
Cleanroom Pressurization Control	• Cleanroom Pressure Control shall be achieved via the Direct Pressure-Differential Control (DP) method. A pressure differential sensor measures the pressure difference between the Cleanroom and adjacent space (outside ambient, via the ZRP). • The Makeup Air Unit fan speeds shall be automatically modulated, to adjust the Makeup airflow amount supplied to the Cleanroom, to achieve the required pressure differential.

<u>Note:</u> The control philosophy for the MAUs, Exhaust, and Chilled Water Systems will not be covered in this Standard.

16.2 The following table shows the control valve fail positions that shall be configured. See **Table 26**.

Table 26: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
Dry Cooling Coil Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State

16.3 Any control valve actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

16.4 Upon a loss event of control air, power, or PLC signal, and then after the utility is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

16.5 Control valves shall have a manual override bypass, to allow manual operation of the valve upon a failure.

16.6 SCADA Monitoring and Alarms: The following SCADA Alarms shall be enabled for Probe Cleanroom Temperature and Humidity:

- HighHigh: 24.5°C and 50.0% RH
- High: 24.0°C and 48.0% RH
- Low: 21.0°C and 42.0% RH
- LowLow: 20.5°C and 40.0% RH

16.7 On the SCADA graphics screen for Cleanroom Temperature and Humidity, each T&H transmitter location shall also show the equivalent Dewpoint Temperature in SCADA.

17.0 Inter-Discipline Coordination Matrix:

17.1 The following table shows the provisions that shall be provided by each of the following disciplines and coordinated accordingly. See **Table 27**.

Table 27: Provisions Provided by Each Discipline

Discipline	Provision
Architectural	• Cut openings in walls and floors for piping penetrations, and seal openings after piping installation • Provide adequate fire stopping at all required fire wall penetrations There shall be no air infiltration or leakage from the outside spaces into the Cleanroom including the Return Air Chase or Interstitial Plenum. This includes from, but not limited to, the following measures: • Building construction seams, joints, and gaps shall be covered with separation panels and sealed to prevent infiltration and leakage. • All joints in walls, floors, and ceilings shall be of vapor-tight construction. • Sealing and caulking with cleanroom approved sealant shall be used to seal all gaps between panels and gaps in cut-outs.
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized equipment. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide Makeup Air Unit(s), heating water and chilled water supply piping, supply ductwork, and accessories as outlined in this Standard • Provide Exhaust Fan(s), exhaust ductwork, and accessories as outlined in this Standard • Provide chilled water piping and accessories as outlined in this Standard • Provide floor drains located suitably to allow proper draining of equipment, condensate drains and auto-drains, piping, and air vents

Discipline	Provision
Electrical	• Provide power wiring for all Electrical equipment/components • Provide grounding for all Electrical equipment/components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide drain source for the Condensate Drain water piping
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, dewpoint, flow, etc. • Provide Zero Reference Pipe (ZRP) for pressure reference • Integrate FFU System Controller with FMCS/SCADA

18.0 Quality Inspection and Testing:

18.1 **Inspection and Test Plan (ITP)** – Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

18.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping, Ductwork, and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning, Flushing, and Purging
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor's Procedure to allow equipment to run in MANUAL mode. Also considered "Early Run" Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

18.2 **Cleanroom QA/QC Inspection** – Prior to testing, QA/QC inspection shall be performed on all installed equipment, piping, ductwork, and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

19.0 Startup Commissioning / Site Acceptance Test

19.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test.

19.1.1 Cleanroom Temperature

- Verify Dry Cooling Coil CHW Control Valve Function through the entire operating range (0 to 100%)
- Confirm Dry Cooling Coil CHW Control Valve control in response to Cleanroom Dry-bulb Temperature fluctuations

19.1.2 Cleanroom Humidity

- Confirm Makeup Air Unit discharge setpoint control in response to Cleanroom Dewpoint Temperature fluctuations

19.1.3 Cleanroom Pressurization

- Confirm Makeup Air Unit fan speed setpoint control in response to Cleanroom Pressure fluctuations

19.1.4 Cleanroom FFUs

- Confirm FFU fan speed setpoint and speed adjustment control through the entire operating range (0 to 100%)

19.1.5 Instrumentation and Controls

- Confirm Auto-switch to backup Cleanroom Temperature & Humidity transmitter for a single Temperature Control Zone, upon failure of main Cleanroom Temperature & Humidity transmitter in that zone
- Confirm Auto-switch to the other backup Cleanroom Pressure transmitters, upon failure of one Cleanroom Pressure transmitter
- Confirm all control valves fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA

20.0 Reference Specifications:

20.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 28**.

Table 28: Reference Construction Specifications

Spec Section	Spec Contents	Remarks
Section 13 21 13	Clean Rooms	
Section 09 58 13	Cleanroom Gasketed Ceiling System	
Section 09 69 60	Cleanroom Access Flooring	
Section 10 22 20	Cleanroom Partitions	
Section 10 57 53	Cleanroom Specialties	
Section 23 34 40	Cleanroom Fan Filter Units	
Section 21 10 00	Automatic Sprinkler System	
Section 23 77 03	Cleanroom Makeup Air Handling Units	
Section 23 41 00	Air Filters, HVAC	

21.0 Document Control

21.1 **Approval**: GLOBAL_FAC_GFTT_APPR

21.2 **Notification**:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

21.3 **Final Viewer Access**

All Micron Team Members at all Sites

21.4 **Retention**

Adherence to the requirements specified in **Global Facilities - Document Control Procedures**.

21.5 **Review**

Biennially (two years)

22.0 Revision History

Revision #	**Section Changed**	**Details of Changes**	**Date**
0	NA	New document	08/29/2023

Appendix 1: Template Form for Site Specific Deviation List

Site:______________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT